PACKET SNIFFING USING RAW SOCKETS

Aim: To capture packets and display packet-header information using raw sockets.

Program:

- raw-sniffer.py

```
import socket
import struct
import sys
def main():
try:
  sniffer = socket.socket(socket.AF_INET, socket.SOCK_RAW, socket.IPPROTO_IP)
  except OSError as exe:
  print("Could not create raw socket:", exe)
  print("On linux, try running with sudo on an authorized test machine.")
  return
  print("Capturing IP packets. Press Ctrl+C to stop.")
try:
  while True:
    packet, address = sniffer.recvfrom(65535)
    if len(packet) < 20:
    continue
    version_ihl = packet[0]
    version = version_ihl >> 4
    ihl = (version_ihl & 0x0F) * 4
    total_length = struct.unpack("!H", packet[2:4])[0]
    source = socket.inet_ntoa(packet[16:20])
  if version == 4 and ihl >= 20:
      print("-" * 50)
      print("Source IP:", source)
      print("Destination IP:", destination)
      print("Protocol number:", protocol)
      print("IP header size:", ihl, "bytes")
      print("Total length:", total_length, "bytes")
except KeyboardInterrupt:
    print("\n Capture stopped.")
finally:
    sniffer.close()
if __name__ == "__main__":
    main()
```

Results:

Packet-header information was captured and displayed in a authorized test environment.

Output:

Source IP : 192.168.1.10

Destination IP : 192.168.1.1

Protocol Number : 6

IP Header Size : 20 bytes

Total length : 1500 bytes

Source IP : 192.168.1.10

Destination IP : 192.168.1.10

Protocol Number : 17

IP Header Size : 20 bytes

Total length : 60 bytes

^C

captured Stopped.



